ORIENTATION TO COMPUTING

UNIT 5: IP ADDRESSING, SUBNETTING, ROUTING & CORE INTERNET PROTOCOLS

Curriculum: Lovely Professional University | School of Computer Science & Engineering
Reference: Data Communications and Networking with TCP/IP Protocol Suite by Behrouz A. Forouzan

>> 5.1 IPv4 Addressing Architecture and Classful Allocation

An Internet Protocol (IP) address is a 32-bit unique numerical identifier assigned to every network interface card (NIC) connected to an IP network:
* Dotted-Decimal Notation: 32 bits divided into four 8-bit octets separated by periods (e.g., 192.168.1.1). Range of each octet: 0 to 255.
* Original Classful Addressing Scheme (RFC 791):
- Class A: First bit 0 (Range: 1.0.0.0 to 126.255.255.255). 8 network bits, 24 host bits. 126 networks, 16.7 million hosts each. Default mask: 255.0.0.0 (/8).
- Class B: First bits 10 (Range: 128.0.0.0 to 191.255.255.255). 16 network bits, 16 host bits. 16,384 networks, 65,534 hosts each. Default mask: 255.255.0.0 (/16).
- Class C: First bits 110 (Range: 192.0.0.0 to 223.255.255.255). 24 network bits, 8 host bits. 2 million networks, 254 hosts each. Default mask: 255.255.255.0 (/24).
- Class D: First bits 1110 (Range: 224.0.0.0 to 239.255.255.255). Reserved for Multicast group addressing.
- Class E: First bits 1111 (Range: 240.0.0.0 to 255.255.255.255). Reserved for experimental research.
* Loopback Address: 127.0.0.1 (localhost) used for internal software diagnostics.
* Private IP Ranges (RFC 1918 - non-routable on public Internet):
- Class A: 10.0.0.0 to 10.255.255.255 (10.0.0.0/8)
- Class B: 172.16.0.0 to 172.31.255.255 (172.16.0.0/12)
- Class C: 192.168.0.0 to 192.168.255.255 (192.168.0.0/16)

>> 5.2 Subnetting, CIDR and Variable Length Subnet Masking (VLSM)

* Subnetting: Borrowing host bits to create smaller, manageable subnetworks to reduce broadcast traffic and enhance routing security.
* Classless Inter-Domain Routing (CIDR): Suffix notation (/N) indicating number of contiguous masked network bits.
* Formulae for Subnet Calculations:

- Number of Subnets created = 2^b (where b = borrowed bits from host field).
- Total IP addresses per subnet = 2^h (where h = remaining host bits).
- Usable host addresses per subnet = 2^h - 2 (subtracting Network ID where all host bits = 0, and Directed Broadcast ID where all host bits = 1).
* Variable Length Subnet Masking (VLSM): Allocating subnets with differing prefix lengths tailored to exact departmental host requirements, completely eliminating IP address waste.

>> 5.3 IPv6 Protocol Architecture

* Motivation: Complete exhaustion of IPv4 32-bit address space (4.3 billion addresses).
* Specifications: 128-bit address length written as eight 16-bit hexadecimal blocks separated by colons (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334).
* Total IPv6 Address Space: 2^128 approx 3.4 x 10^38 addresses (virtually inexhaustible).
* Zero Compression Rules: Leading zeros in a block can be dropped (e.g., 0000 becomes 0); consecutive blocks of zeros can be replaced ONCE by a double colon (::).
* Architectural Enhancements: Simplified 40-byte fixed header, elimination of broadcast (replaced by multicast/anycast), mandatory IPSec security, Stateless Address Autoconfiguration (SLAAC).

>> 5.4 Transport Layer Protocols: TCP vs UDP

* Transmission Control Protocol (TCP): Connection-oriented, highly reliable, guaranteed ordered byte-stream delivery. Features: 3-way handshake (SYN, SYN-ACK, ACK), cumulative acknowledgments, sequence numbers, flow control (sliding window), congestion control (Slow Start, Congestion Avoidance, Fast Retransmit).
* User Datagram Protocol (UDP): Connectionless, lightweight, unreliable, low-overhead datagram service. No handshakes, no acknowledgments, no ordering. Used for real-time applications: VoIP, live video streaming, DNS lookups, online multiplayer gaming.

>> 5.5 Core Application and Support Protocols

* DNS (Domain Name System - UDP/TCP Port 53): Distributed hierarchical database translating human-friendly domain names (e.g., lpu.in) into machine routable IP addresses.
* DHCP (Dynamic Host Configuration Protocol - UDP Ports 67/68): Automatically assigns IP, subnet mask, default gateway, and DNS servers via the 4-step DORA process

(Discover, Offer, Request, Acknowledge).
* ARP (Address Resolution Protocol): Resolves an unknown Layer 2 physical MAC address corresponding to a known Layer 3 logical IP address on a local broadcast domain.
* HTTP / HTTPS (Ports 80 / 443): Hypertext Transfer Protocol. HTTPS encapsulates HTTP within an encrypted SSL/TLS tunnel.

[PROBLEM 1: Subnetting a Class C Network into 4 Subnets]

Problem Statement: An academic department is assigned the Class C network address 192.168.10.0/24. Subnet this network into 4 equal subnets. For each subnet, determine: (1) Subnet Mask in dotted-decimal and CIDR notation, (2) Subnet Network ID, (3) Usable Host IP Range, and (4) Directed Broadcast Address.
Step-by-Step Mathematical Derivation:
* 1. Determine Borrowed Bits (b):
- Number of required subnets = 4 -> 2^b >= 4 -> b = 2 bits.
* 2. Calculate New Subnet Mask:
- Original mask = /24 (24 bits). New mask = 24 + 2 = /26 bits.
- 4th octet binary = 11000000_2 = 128 + 64 = 192.
- Subnet Mask = 255.255.255.192 (/26).
* 3. Calculate Block Size (Delta):
- Remaining host bits h = 8 - 2 = 6 bits. Block size = 2^6 = 64 (or 256 - 192 = 64).
* 4. Detailed Subnet Allocation Table:
- Subnet 1: Network ID = 192.168.10.0/26 | Usable Range: 192.168.10.1 to 192.168.10.62 | Broadcast: 192.168.10.63.
- Subnet 2: Network ID = 192.168.10.64/26 | Usable Range: 192.168.10.65 to 192.168.10.126 | Broadcast: 192.168.10.127.
- Subnet 3: Network ID = 192.168.10.128/26 | Usable Range: 192.168.10.129 to 192.168.10.190 | Broadcast: 192.168.10.191.
- Subnet 4: Network ID = 192.168.10.192/26 | Usable Range: 192.168.10.193 to 192.168.10.254 | Broadcast: 192.168.10.255.
* 5. Verification: Total usable hosts = 4 x 62 = 248 hosts.

[PROBLEM 2: Variable Length Subnet Masking (VLSM) Enterprise Design]

Problem Statement: Design an optimal VLSM address plan for network 172.16.0.0/22 for four departments with the following host requirements: Engineering (400 hosts), Sales (120 hosts), Admin (50 hosts), and WAN Point-to-Point Link (2 hosts).
Step-by-Step VLSM Allocation Protocol (Highest to Lowest):

* 1. Engineering (400 hosts):
- 2^h - 2 >= 400 -> h = 9 (2^9 - 2 = 510 hosts). Mask = 32 - 9 = /23 (255.255.254.0).
- Allocate: 172.16.0.0/23 (Range: 172.16.0.1 - 172.16.1.254; Broadcast: 172.16.1.255).
Next available = 172.16.2.0.
* 2. Sales (120 hosts):
- 2^h - 2 >= 120 -> h = 7 (2^7 - 2 = 126 hosts). Mask = 32 - 7 = /25 (255.255.255.128).
- Allocate: 172.16.2.0/25 (Range: 172.16.2.1 - 172.16.2.126; Broadcast: 172.16.2.127).
Next available = 172.16.2.128.
* 3. Admin (50 hosts):
- 2^h - 2 >= 50 -> h = 6 (2^6 - 2 = 62 hosts). Mask = 32 - 6 = /26 (255.255.255.192).
- Allocate: 172.16.2.128/26 (Range: 172.16.2.129 - 172.16.2.190; Broadcast:
172.16.2.191). Next available = 172.16.2.192.
* 4. WAN Link (2 hosts):
- 2^h - 2 >= 2 -> h = 2 (2^2 - 2 = 2 hosts). Mask = 32 - 2 = /30 (255.255.255.252).
- Allocate: 172.16.2.192/30 (Range: 172.16.2.193 - 172.16.2.194; Broadcast:
172.16.2.195).

[PROBLEM 3: TCP 3-Way Handshake Connection Sequence Analysis]

Problem Statement: Trace the sequence numbers, acknowledgment numbers, and TCP control flags during the establishment and termination of a reliable connection between Client C and Web Server S.
Step-by-Step TCP Protocol Sequence:
* 1. Handshake Step 1 (SYN from Client):
- Client sends SYN packet to Server with an Initial Sequence Number (ISN_c = 5000).
Control flag: SYN = 1, ACK = 0.
* 2. Handshake Step 2 (SYN-ACK from Server):
- Server receives SYN, allocates buffer resources, and replies with SYN-ACK. Server selects its own Initial Sequence Number (ISN_s = 8000). Acknowledgment Number = ISN_c + 1 = 5001. Control flags: SYN = 1, ACK = 1.
* 3. Handshake Step 3 (ACK from Client):
- Client replies with ACK packet. Sequence Number = 5001. Acknowledgment Number = ISN_s + 1 = 8001. Control flags: SYN = 0, ACK = 1.
- Connection state transitions to ESTABLISHED on both endpoints. Data transfer commences.
* 4. Connection Teardown (4-Way Handshake):
- Client sends FIN (FIN=1, ACK=1). Server replies with ACK. (Half-closed: server can still send remaining data).
- Server completes transmission and sends FIN. Client replies with final ACK and enters TIME_WAIT state (2 x MSL = 120 s) to ensure ACK reaches server safely before closing

socket.

[PROBLEM 4: Walkthrough of DHCP DORA and DNS Iterative Resolution]

Problem Statement: Trace the exact sequence of network protocol packets when an unconfigured laptop connects to an enterprise campus Wi-Fi network and requests webpage "www.lpu.in".

End-to-End Network Protocol Flow:

* 1. DHCP Discover: Laptop broadcasts UDP packet from 0.0.0.0:68 to 255.255.255.255:67 with its MAC address.
* 2. DHCP Offer: Campus DHCP server reserves IP 10.20.30.45 and offers it to client along with subnet mask 255.255.255.0, default gateway 10.20.30.1, and DNS server 10.20.1.10.
* 3. DHCP Request: Laptop broadcasts formal request to lease offered IP 10.20.30.45.
* 4. DHCP ACK: Server confirms lease in binding database; laptop configures network adapter.
* 5. ARP Resolution: Laptop checks if default gateway MAC is cached; sends ARP request for 10.20.30.1 to obtain gateway MAC.
* 6. DNS Resolution: Laptop sends UDP query to DNS server (10.20.1.10) for "www.lpu.in".
* 7. Iterative DNS Query Hierarchy: Local DNS queries Root Server (.) -> TLD Server (.in) -> Authoritative Nameserver (lpu.in) -> Returns IP 14.139.242.10 to laptop.
* 8. TCP Connection: Laptop initiates TCP 3-Way Handshake with 14.139.242.10 on port 443 (HTTPS).